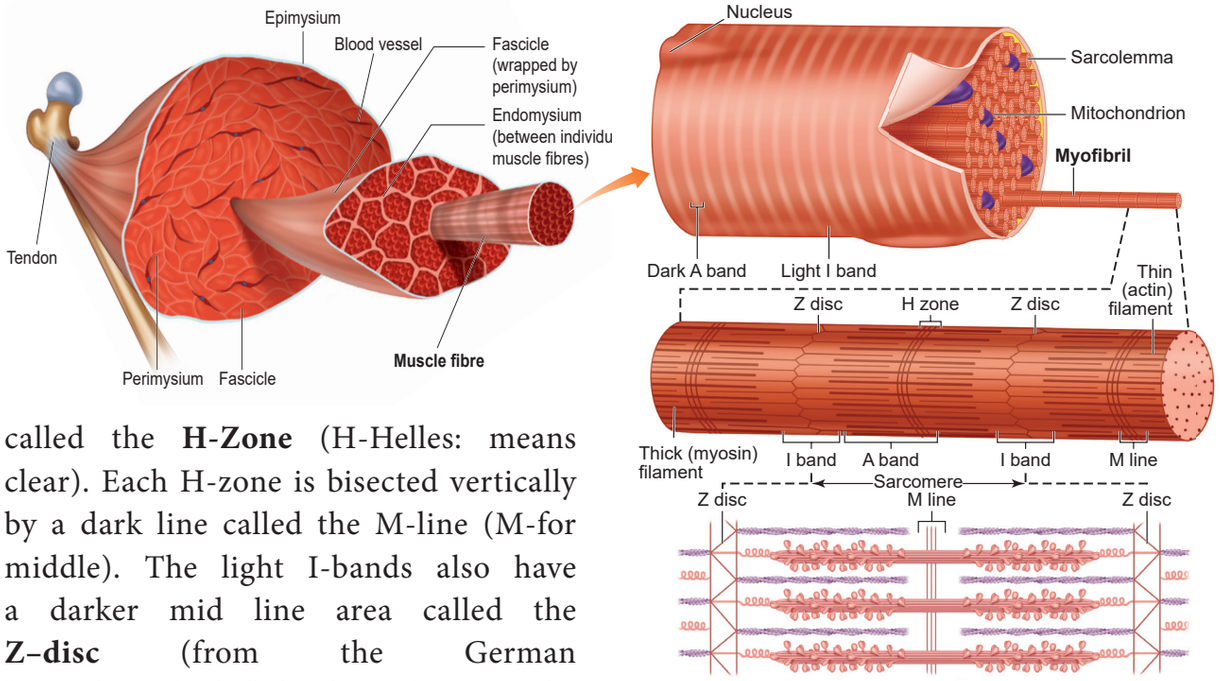



**Figure 9.1** Organizational level of a skeletal muscle

called the **H-Zone** (H-Helles: means clear). Each H-zone is bisected vertically by a dark line called the M-line (M-for middle). The light I-bands also have a darker mid line area called the **Z–disc** (from the German "Zwischenscheibe" the disc inbetween the I-bands).

The myofibrils contain the contractile element, the **sarcomere** which is the functional unit of the skeletal muscle. A Sarcomere is the region of a myofibril between two successive Z-discs. It contains an A-band with a half I-band at each end. Inside the sarcomere two types of filaments are present namely the **thick** and **thin filaments**.

The thick filaments extend the entire length of the A-band, the thin filaments extend across the I-band and partly into the A-band. The invagination of the sarcolemma forms transverse tubules **(T–tubules)** and they penetrate into the junction between the A and I-bands.

**Muscle Terminology**

| General Term | Muscle Equivalent |
|---|---|
| Cell | Muscle fibre/ Myofibril |
| Plasma membrane | Sarcolemma |
| Cytoplasm | Sarcoplasm |
| Endoplasmic reticulum | Sarcoplasmic reticulum |

## 9.4 Structure of contractile proteins

Contraction of the muscle depends on the presence of contractile proteins (Figure 9.2) such as **actin** and **myosin** in the myofilaments. The thick filaments are composed of the protein myosin which are bundled together whose heads produce at opposite ends of the filament. Each myosin molecule is made up of a monomer called meromyosin. The meromyosin has two regions, a globular head with a short arm and a tail. The short arm constitutes the heavy meromyosin (HMM). The tail portion forms the light meromyosin (LMM). The head bears an actin-binding site and an ATP- binding site. It also contains ATPase enzyme that split ATP to generate energy for the contraction of muscle. The thin filaments are composed of two interwined actin molecules. Actin has polypeptide subunits called globular



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